# INORGANIC CUME
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**HinderedAxial-Equatorial Carbonyl Exchange in an $Fe(CO)_4(PR_3)$ Complex of a Rigid Bicyclic Phosphine** **Thomas S. Barnard and Mark R. Mason***

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1) The title of the paper in question suggests that equatorial-axial ligand exchange is rapid for most trigonal bipyramid complexes. What is the molecular basis that explains this facile exchange?

2) The four phosphine ligands shown in the Appendix were used to prepare Fe(CO)4L [L = phosphine ligands 1 through 4]. Which of the possible isomeric structures was observed for these complexes?

3) Briefly decribe how the experimental data in the appended information was used to assign ligand characteristics for these phosphines.

4) What inference can be drawn from the data in Table 5 about the differences or similarities in the bonding in $Fe(CO)_4L$? [L = 1 through 4]?

5) For which of the phosphine complexes was slow exchange noted and how was the exchange rate determined?

6) Briefly give your analysis of the probable reason for the case(s) where slow exchange is noted.

7) Why was it important or the authors to collect data on all four $Fe(CO)_4L$ systems before reaching their conclusion about the origin of the slow $CO_{eq}$ - $CO_{ax}$ exchange?